Context

This writing sample is based on an independent experimental study I conducted during my undergraduate program in Computer Science. The project grew out of my interest in understanding how machine learning models behave under distributional imbalance, particularly when standard performance metrics fail to reflect growing misalignment between confidence and correctness. I designed the experiments, implemented the training and evaluation pipelines, and carried out the analysis and writing independently. The work reflects my attempt to reason carefully about model behavior under controlled conditions, rather than to propose new methods or optimizations.

Class Imbalance and Confidence Calibration in Convolutional Neural Networks

1. Abstract

Class imbalance is a common characteristic of real-world datasets, yet its effects are often evaluated primarily through aggregate accuracy metrics. In this work, I present a controlled empirical study examining how class imbalance influences not only predictive performance but also the calibration of convolutional neural networks. Using CIFAR-10 with systematically induced imbalance, I show that while overall test accuracy degrades moderately, minority classes exhibit severe miscalibration, characterized by persistent overconfidence in incorrect predictions. Through reliability diagrams and Expected Calibration Error (ECE), I demonstrate that majority classes remain comparatively well-calibrated, masking substantial failures on underrepresented classes. I further evaluate post-hoc temperature scaling as a calibration control and find that, although it improves global calibration, it does not eliminate minority-class miscalibration. These results highlight a critical limitation of standard evaluation and calibration practices under class imbalance and underscore the need to consider confidence behavior alongside accuracy when deploying neural networks in imbalanced settings.

2. Introduction

Class imbalance is a pervasive feature of real-world data, arising naturally in domains such as medical imaging, fault detection, and risk assessment, where rare but critical cases are underrepresented. While modern convolutional neural networks (CNNs) often achieve strong aggregate performance under such conditions, it is increasingly unclear whether traditional evaluation metrics sufficiently capture their behavior in imbalanced settings. In particular, high overall accuracy can coexist with systematic failures on minority classes, raising concerns about the reliability of deployed models.

Existing analyses of class imbalance predominantly emphasize class-wise accuracy degradation and summary metrics such as macro-averaged F1 score. While these metrics provide useful signals, they implicitly assume that incorrect predictions are uniformly harmful. In practice, however, the confidence associated with a prediction is often as important as the prediction itself. A model that is uncertain and wrong may be manageable, whereas a model that is confidently wrong can be actively misleading. This distinction becomes especially important in safety-critical or decision-support systems, where confidence is frequently used to guide downstream actions.

Despite this, comparatively little attention is paid to how class imbalance affects model calibration, the alignment between predicted confidence and empirical correctness. A well-calibrated model should produce confidence estimates that reflect true likelihoods of correctness, yet neural networks are known to exhibit overconfidence even under balanced training conditions. Whether and how this behavior changes under class imbalance remains insufficiently characterized, particularly at the level of majority versus minority classes.

In this work, I conduct a controlled empirical study to examine the effects of class imbalance on both predictive performance and calibration behavior in CNNs. Using CIFAR-10 as a standardized testbed, I systematically induce class imbalance while holding architecture, optimization, and evaluation procedures fixed. This design allows imbalance to be isolated as the sole experimental variable.

My results show that while aggregate accuracy degrades only moderately under imbalance, calibration failures emerge asymmetrically across classes. Majority classes remain relatively well-calibrated, while minority classes exhibit pronounced overconfidence, leading to a substantial increase in Expected Calibration Error (ECE) and visibly distorted reliability diagrams. I further evaluate post-hoc temperature scaling as a standard calibration technique and demonstrate that, although it improves global calibration metrics, it does not fully correct minority-class miscalibration.

These findings suggest that commonly reported performance metrics can obscure harmful failure modes under class imbalance and that post-hoc calibration alone may be insufficient to address confidence misalignment in underrepresented classes. By highlighting this discrepancy, this study aims to motivate more nuanced evaluation practices that consider both accuracy and confidence behavior when assessing neural networks trained on imbalanced data.

3. Experimental Setup

All experiments in this study are designed to isolate class imbalance as the sole experimental variable. To ensure that observed effects can be attributed directly to imbalance rather than architectural or optimization differences, the network architecture, training procedure, and evaluation protocol are held fixed across all experimental conditions.

Dataset: CIFAR-10 is used as the experimental benchmark. The dataset consists of 60,000 color images across 10 classes, with a standard split of 50,000 training samples and 10,000 test samples. Under the balanced setting, each class contributes an equal number of training examples. To induce class imbalance, a subset of classes is designated as minority classes, and their training samples are uniformly subsampled while all other classes are left unchanged. The test set is kept fully balanced in all experiments to ensure fair and consistent evaluation across conditions.

Imbalance construction: Five classes are treated as minority classes, with their training samples reduced to 20% of the original count. This results in a training distribution where majority classes retain 5,000 samples each, while minority classes contain 1,000 samples each. This setting reflects a moderate but realistic imbalance scenario, allowing degradation effects to be observed without collapsing training altogether. Importantly, no reweighting, resampling, or data augmentation techniques are applied, as the objective of this study is characterization rather than mitigation.

Model architecture: A simple convolutional neural network is used to facilitate interpretability. The network consists of two convolutional layers with ReLU activations and max-pooling, followed by two fully connected layers. Architectural complexity is intentionally limited in order to expose failure modes rather than obscure them through capacity or regularization effects. The same architecture is used for both balanced and imbalanced training.

Training procedure: All models are trained using cross-entropy loss and the Adam optimizer with a fixed learning rate. Training is performed for a small, fixed number of epochs sufficient to achieve convergence without overfitting. Random seeds are fixed across all runs to ensure reproducibility. Aside from the induced class imbalance, no changes are made to the training process between experiments.

Evaluation protocol: Model performance is evaluated on the standard CIFAR-10 test set. In addition to overall accuracy, calibration behavior is assessed using Expected Calibration Error (ECE) and reliability diagrams. To examine asymmetric effects of imbalance, calibration metrics are computed

separately for majority (head) and minority (tail) classes. This disaggregated evaluation enables identification of failure modes that are obscured by aggregate metrics.

Calibration control: To assess whether observed miscalibration can be corrected post hoc, temperature scaling is applied as a calibration control. A single scalar temperature parameter is learned on a held-out validation split and applied to test-set logits without modifying model weights. This procedure allows calibration effects to be evaluated independently of changes in predictive accuracy.

4. Evaluation Metrics

Model performance under class imbalance is commonly evaluated using aggregate accuracy or class-averaged metrics such as macro-F1 score. While these measures provide a useful summary of predictive correctness, they do not capture how confident a model is in its predictions. In many practical settings, predictive confidence plays a critical role in downstream decision-making, and errors made with high confidence can be more harmful than uncertain mistakes. For this reason, evaluation based solely on accuracy can obscure important failure modes.

Predictive performance: Overall test accuracy is reported to quantify aggregate classification performance. Macro-averaged F1 score is additionally computed to account for class-wise performance differences under imbalance. These metrics are included to provide continuity with prior work and to establish baseline performance degradation under class imbalance. However, they are not treated as sufficient indicators of model reliability.

Calibration metrics: To assess confidence alignment, Expected Calibration Error (ECE) is used as the primary calibration metric. ECE measures the discrepancy between predicted confidence and empirical accuracy by partitioning predictions into confidence bins and computing the weighted average of the absolute difference between confidence and accuracy within each bin. Lower ECE values indicate better calibration, while higher values reflect systematic overconfidence or underconfidence.

In addition to scalar ECE values, reliability diagrams are employed to visualize calibration behavior. These plots compare average predicted confidence against empirical accuracy across confidence bins, providing an interpretable depiction of whether a model's confidence estimates reflect its true likelihood of correctness. Reliability diagrams are particularly useful for identifying confidence distortions that may not be evident from summary statistics alone.

Disaggregated evaluation: To examine whether class imbalance affects calibration uniformly across classes, calibration metrics are computed separately for majority (head) and minority (tail) classes. This disaggregation enables direct comparison of confidence behavior between well-represented and underrepresented classes and allows identification of asymmetric failure modes that are masked by aggregate evaluation.

Post-hoc calibration assessment: Finally, temperature scaling is applied as a post-hoc calibration control. Calibration metrics are recomputed after scaling to evaluate whether confidence misalignment induced by class imbalance can be corrected without modifying model parameters. This comparison helps distinguish between calibration errors that are globally correctable and those that persist specifically in minority classes.

5. Results

This section presents the empirical results of the study, organized to progressively examine the effects of class imbalance on predictive performance, calibration behavior, and the effectiveness of post-hoc calibration. Unless otherwise stated, all results are reported on the held-out test set.

5.1 Predictive Performance Under Class Imbalance

To establish a baseline, model performance is first evaluated using standard predictive metrics. Under balanced training, the network achieves stable convergence and competitive test accuracy. When class imbalance is introduced, overall test accuracy decreases, but the magnitude of degradation remains moderate.

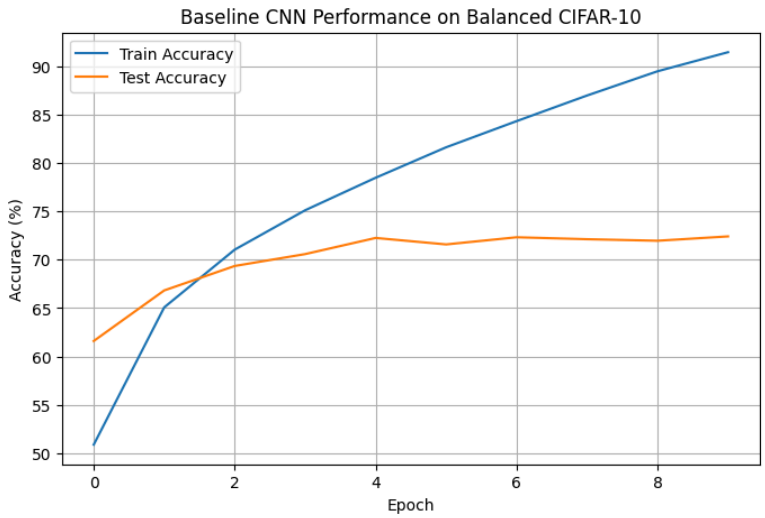

Figure 1: *Training and test accuracy of the CNN on balanced CIFAR-10, demonstrating stable learning behavior and a moderate generalization gap.*

Macro-averaged F1 score exhibits a more pronounced decline, reflecting reduced performance on underrepresented classes. This behavior is consistent with prior observations in imbalanced classification and confirms that minority classes are disproportionately affected in terms of predictive correctness. However, neither overall accuracy nor macro-F1 fully captures how the model's confidence changes under imbalance. In particular, these metrics provide no indication of whether incorrect predictions are made with appropriate uncertainty or excessive confidence.

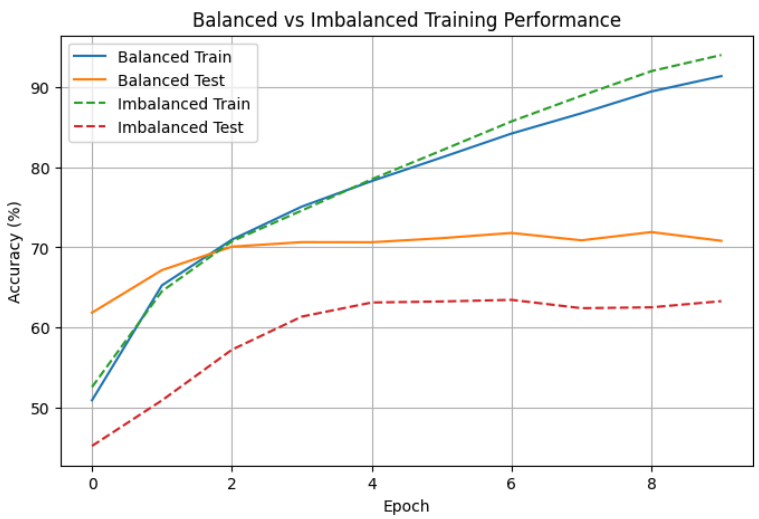

Figure 2: *Comparison of training and test accuracy under balanced and imbalanced training. While imbalanced training achieves high training accuracy, test performance degrades, indicating reduced generalization.*

These results motivate a closer examination of calibration behavior, as predictive performance alone suggests only a partial characterization of the model's failure modes under imbalance.

5.2 Calibration Behavior and Asymmetric Failure Modes

Calibration analysis reveals a substantially different picture from that suggested by predictive metrics alone. Under imbalanced training, Expected Calibration Error (ECE) increases notably relative to the balanced baseline, indicating a systematic mismatch between predicted confidence and empirical accuracy.

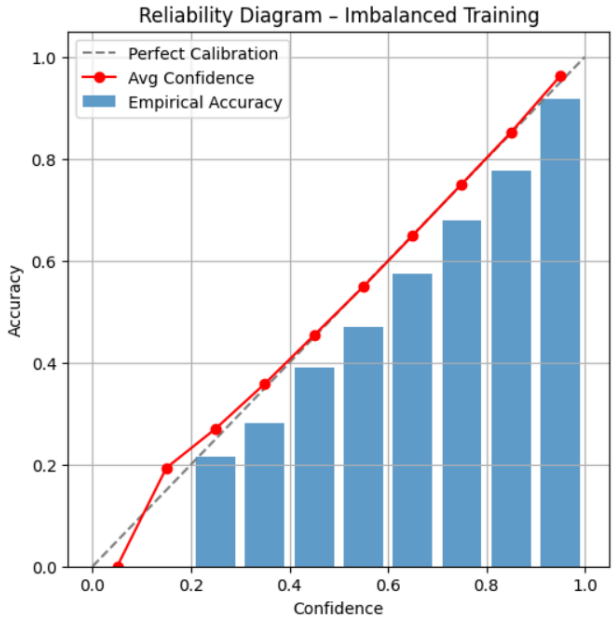

Figure 3: *Reliability diagram of the imbalanced-trained model on the test set.*

Reliability diagrams further illustrate this effect. While the model continues to assign high confidence to many predictions, empirical accuracy within these confidence bins falls short of the predicted confidence, particularly at higher confidence levels. This pattern indicates overconfidence rather than random uncertainty and suggests that the model's confidence estimates no longer reliably reflect its true likelihood of correctness.

To assess whether this miscalibration affects all classes uniformly, calibration metrics are computed separately for majority (head) and minority (tail) classes. This disaggregated evaluation reveals a pronounced asymmetry. Majority classes remain relatively well-calibrated, exhibiting low ECE and reliability curves that closely track the ideal diagonal. In contrast, minority classes exhibit substantially higher ECE and visibly distorted reliability diagrams, with confidence consistently exceeding empirical accuracy.

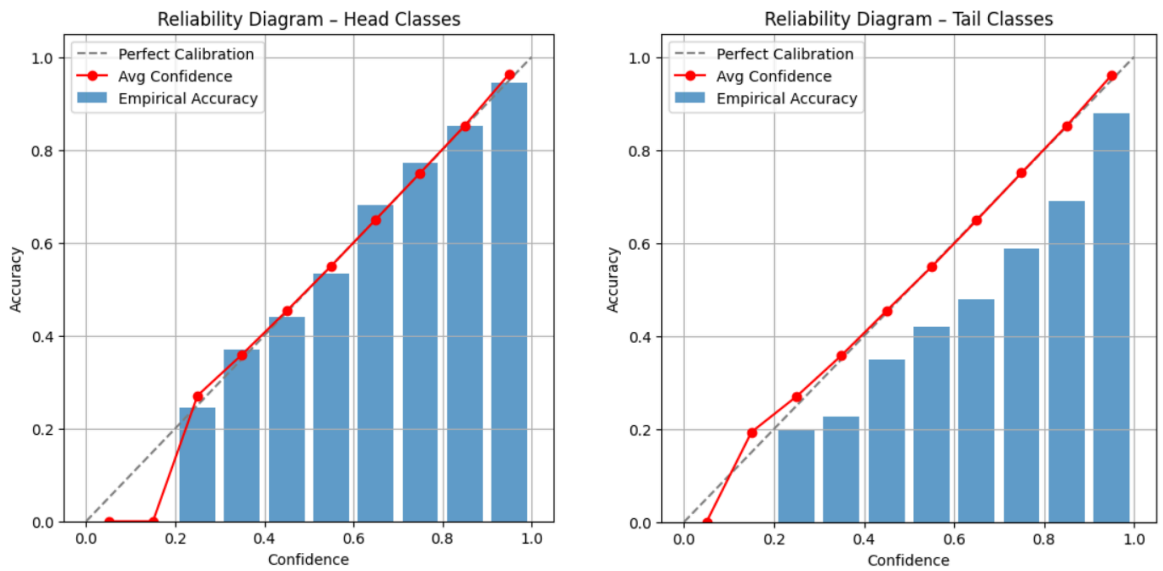

Figure 4: *Reliability diagrams for majority (head) and minority (tail) classes under imbalanced training.*

Notably, this asymmetric behavior is largely obscured when calibration is evaluated in aggregate. Because majority classes dominate the test set, their comparatively well-calibrated behavior masks the severe miscalibration present in minority classes. As a result, global calibration metrics understate the extent of confidence misalignment affecting underrepresented classes.

5.3 Effect of Post-hoc Temperature Scaling

To evaluate whether the observed miscalibration can be corrected without modifying model parameters, temperature scaling is applied as a post-hoc calibration control. A single temperature parameter is learned on a held-out validation split and applied to test-set logits.

After temperature scaling, overall calibration improves, as reflected by a reduction in global ECE. Calibration of majority classes also improves modestly, remaining well-aligned with empirical accuracy. These results indicate that temperature scaling is effective at correcting global overconfidence and smoothing confidence estimates across the model's output distribution.

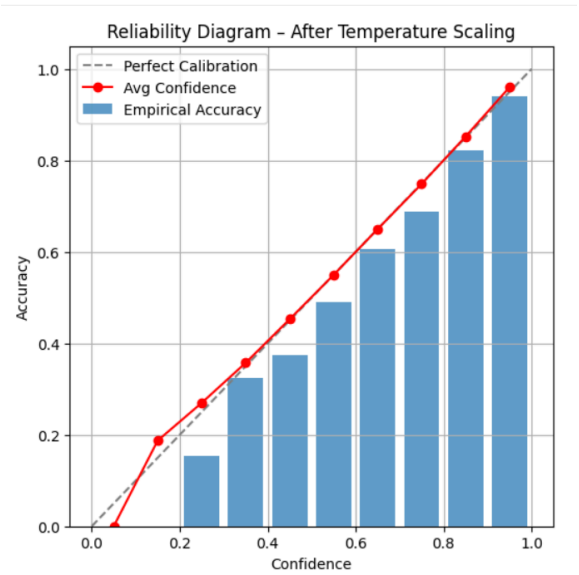

Figure 5: *Reliability diagram after applying post-hoc temperature scaling.*

However, minority-class calibration remains substantially worse than that of majority classes even after scaling. While tail-class ECE decreases slightly, it remains significantly higher than head-class ECE, and reliability diagrams continue to show systematic overconfidence at moderate to high confidence levels. This persistence suggests that minority-class miscalibration is not fully attributable to a global confidence scaling issue, but rather reflects deeper uncertainty induced by data scarcity.

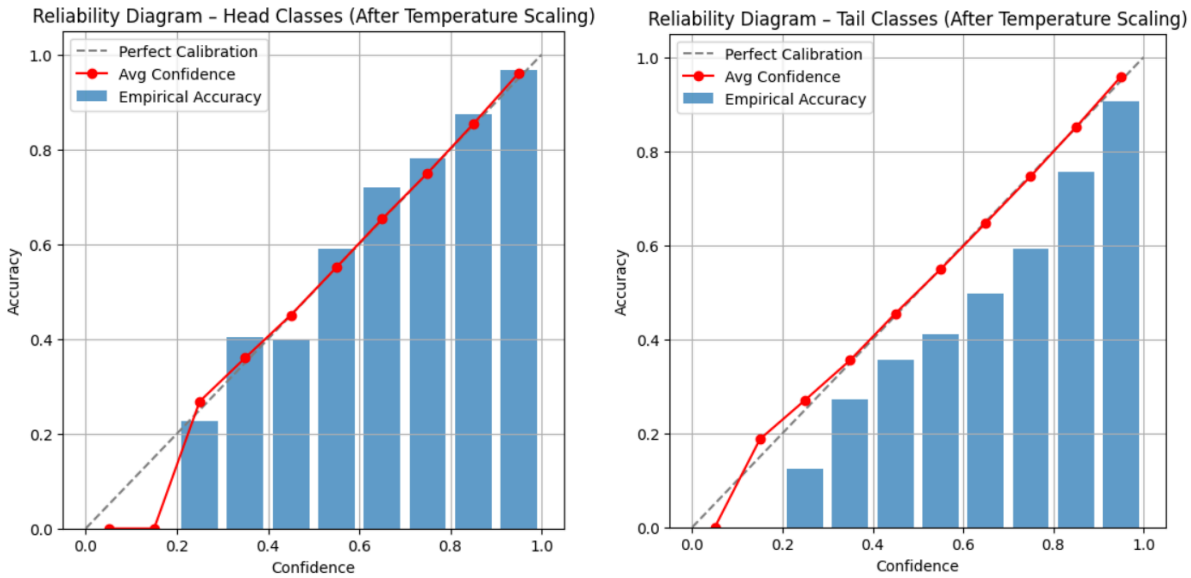

Figure 6: *Reliability diagrams for majority (head) and minority (tail) classes after temperature scaling.*

Together, these findings indicate that post-hoc calibration improves overall confidence alignment but does not eliminate the asymmetric calibration failures introduced by class imbalance.

6. Analysis and Discussion

This study examined the effects of class imbalance not only on predictive performance, but also on the reliability of confidence estimates produced by a convolutional neural network. While class imbalance is often discussed primarily in terms of reduced accuracy on minority classes, the results here highlight a more subtle and potentially more consequential failure mode: systematic miscalibration that disproportionately affects underrepresented classes.

A key observation is that standard predictive metrics provide an incomplete picture of model behavior under imbalance. Although overall test accuracy degrades under imbalanced training, the magnitude of this degradation remains moderate. Macro-averaged F1 score reveals a sharper decline, confirming that minority classes are adversely affected. However, neither metric captures how confidently the model makes its predictions. As a result, a model may appear to generalize reasonably well while simultaneously exhibiting misleading confidence estimates, particularly for rare classes.

Calibration analysis reveals that this confidence distortion is not uniform across classes. When evaluated in aggregate, miscalibration appears moderate, and reliability diagrams suggest a manageable degree of overconfidence. However, disaggregating calibration by class frequency exposes a pronounced asymmetry. Majority classes remain relatively well-calibrated, with confidence estimates that closely track empirical accuracy. In contrast, minority classes exhibit persistent

overconfidence across a wide range of confidence levels. This behavior is largely masked by aggregate metrics due to the dominance of majority-class samples in the evaluation set.

The results after applying temperature scaling further clarify the nature of this asymmetry. Temperature scaling effectively reduces global miscalibration, as evidenced by lower overall Expected Calibration Error and improved alignment in the aggregate reliability diagram. This indicates that a significant component of the model's overconfidence can be corrected through a simple post-hoc adjustment. However, this improvement is not evenly distributed. While majority-class calibration improves or remains stable, minority-class calibration remains substantially worse even after scaling. The persistence of this gap suggests that minority-class miscalibration is not solely a consequence of global overconfidence, but reflects deeper uncertainty arising from data scarcity during training.

These findings have important implications for the deployment of neural networks in settings where confidence estimates are used for downstream decision-making. In applications such as medical diagnosis, autonomous systems, or risk-sensitive classification, overconfident errors on rare cases may be more harmful than uniformly uncertain predictions. The results suggest that relying on aggregate calibration metrics or post-hoc calibration alone may provide a false sense of reliability, particularly when class imbalance is present.

More broadly, this work underscores the importance of evaluating calibration in a class-conditional manner rather than solely at the dataset level. While the experimental setup is intentionally simple, the observed asymmetry highlights a structural issue that is likely to persist in larger models and real-world datasets. Addressing such imbalance-induced miscalibration may require training-time interventions, such as class-aware losses or uncertainty-aware learning objectives, rather than purely post-hoc adjustments.

7. Conclusion

This work examined the effects of class imbalance on convolutional neural networks from both a predictive and a calibration perspective. While standard evaluation metrics indicate a moderate degradation in accuracy under imbalance, a more detailed analysis reveals that these metrics obscure important confidence-related failure modes. In particular, minority classes exhibit persistent overconfidence, even when overall performance appears acceptable.

By analyzing calibration behavior using Expected Calibration Error and reliability diagrams, this study shows that miscalibration under imbalance is strongly asymmetric across classes. Majority classes remain relatively well-calibrated, whereas minority classes experience systematic confidence misalignment that is masked by aggregate evaluation. Although post-hoc temperature scaling improves global calibration, it does not fully resolve this asymmetry, suggesting that imbalance-induced uncertainty cannot be addressed through global confidence rescaling alone.

Together, these findings highlight the importance of evaluating confidence behavior alongside predictive performance, especially in imbalanced settings where errors on rare classes may carry disproportionate consequences. Rather than proposing a new mitigation strategy, this work aims to clarify where and how existing evaluation practices fall short, and to motivate more careful consideration of calibration when deploying neural networks trained on imbalanced data.

8. Limitations and Future Work

This study is intentionally scoped to isolate and analyze the effects of class imbalance on calibration behavior under controlled conditions. As a result, several limitations should be acknowledged. First,

the experimental setup uses a relatively simple convolutional neural network and a single benchmark dataset. While this simplicity aids interpretability and helps expose calibration failure modes, it limits the extent to which the observed behavior can be directly generalized to deeper architectures or more complex, real-world datasets.

Second, class imbalance is introduced through uniform subsampling of selected classes. This approach provides a clean and reproducible imbalance setting, but does not capture all forms of imbalance encountered in practice, such as long-tailed distributions with varying degrees of rarity or label noise correlated with class frequency. Additionally, the study focuses on post-hoc calibration through temperature scaling, which applies a single global adjustment and therefore cannot account for class-specific or instance-specific uncertainty.

Future work could extend this analysis in several directions. Evaluating calibration behavior under more realistic long-tailed distributions and across different model architectures would help assess the robustness of the observed asymmetry. Beyond post-hoc calibration, training-time approaches such as class-aware losses or uncertainty-aware learning objectives may offer more effective ways to address minority-class miscalibration. Finally, examining how these calibration failures interact with downstream decision-making systems could provide further insight into the practical consequences of overconfident errors in imbalanced settings.

9. Appendix: Experimental Validation and Failure Analysis

This appendix documents an early experimental failure encountered during the course of this study and the steps taken to diagnose and correct it. The purpose of including this analysis is not to highlight negative results, but to provide transparency regarding the experimental process and to establish confidence that the final reported findings reflect genuine model behavior under class imbalance rather than artifacts of an erroneous training setup.

By explicitly validating the experimental pipeline, this appendix serves as a methodological safeguard, ensuring that conclusions drawn in the main body of the paper are well-founded.

9.1 Motivation for Failure Analysis

During initial experimentation with class-imbalanced training regimes, the observed model behavior raised concerns regarding the validity of the results. Across multiple imbalance settings, both training and test accuracy remained near constant and close to chance-level performance. Moreover, these metrics exhibited little to no sensitivity to changes in imbalance severity.

Given that CIFAR-10 contains ten classes, such behavior corresponds to near-random guessing and contradicts both theoretical expectations and prior empirical findings in imbalanced classification. Because this failure mode could invalidate any downstream analysis, a systematic investigation was undertaken before proceeding further.

9.2 Observed Failure Mode

Under the initial experimental configuration, the convolutional neural network failed to learn discriminative representations. Training accuracy remained in the range of approximately 10–15%, while test accuracy hovered near 10% regardless of the minority-class retention ratio.

More critically, increasing or decreasing the severity of class imbalance had no measurable effect on performance. This invariance suggested that the model was not meaningfully responding to changes in data distribution, indicating either a breakdown in the learning signal or an error in the data handling or training logic.

To further characterize this behavior, a confusion matrix and class-wise classification report were generated for the imbalanced training regime. These analyses revealed extreme prediction collapse: the model assigned the vast majority of test samples to a small subset of classes while producing zero or near-zero recall for most categories. Macro-averaged F1 scores were correspondingly close to zero, confirming the degenerate nature of the learned solution.

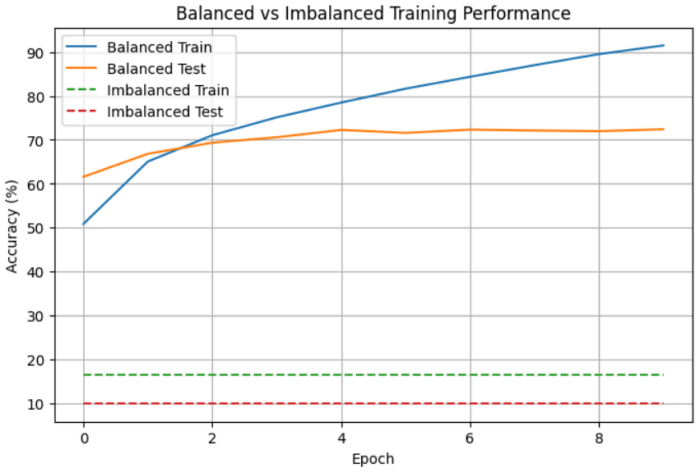

Figure 7: *Training and test accuracy under the initial experimental configuration, showing near-random performance and lack of sensitivity to class imbalance severity.*

9.3 Diagnostic Analysis

To identify the source of this failure, a series of diagnostic checks were conducted:

- **Training dynamics:** Loss trajectories and parameter updates were examined to verify that optimization was occurring and that gradients were being propagated.
- **Data integrity:** Label distributions for both training and test sets were inspected to confirm correct dataset construction under imbalance.
- **Evaluation logic:** Metric computation and data loaders were reviewed to rule out mismatches between inputs, labels, and predictions.
- **Sensitivity checks:** Performance was monitored across multiple imbalance ratios to test whether any variation in behavior emerged.

Collectively, these checks indicated that the observed failure was not due to model capacity limitations, but rather to an error in the data handling and training configuration used during early experimentation. This configuration suppressed effective learning and caused convergence to a trivial solution.

9.4 Corrective Measures and Re-Execution

After identifying the issue, the data sampling and training logic were corrected, and all experiments were rerun from a clean runtime state with fresh model initialization. Particular care was taken to ensure that class imbalance was introduced only through controlled subsampling and that training and evaluation pipelines were consistent.

Following these corrections, training accuracy increased progressively across epochs, and test accuracy rose above the random baseline. Importantly, model performance began to vary

meaningfully with changes in minority-class retention, aligning with theoretical expectations for imbalanced learning.

Under the corrected setup, models trained on increasingly imbalanced data continued to achieve high training accuracy but exhibited degraded generalization performance, particularly for minority classes. This behavior confirmed that the earlier failure was an artifact of the experimental configuration rather than an inherent limitation of the model or dataset.

9.5 Validation Outcome

This failure analysis serves as an important validation step for the study as a whole. By identifying and resolving an early experimental issue, the final results presented in the main paper can be interpreted with greater confidence as reflecting true model behavior under class imbalance.

More broadly, this process highlights the importance of sanity checks, class-wise evaluation, and careful verification when studying deep learning systems in imbalanced data regimes. The inclusion of this appendix ensures that the reported findings are robust, reproducible, and not the result of unintended implementation errors.